ADVANCED SECONDARY PHYSICS

SCENARIO ITEM BANK

FOR CONSTRUCT 1 (FORCE & MOTION)

COMPILED BY

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ASSESSMENT OBJECTIVES, CONSTRUCTS AND THE TOPICS IN THE SYLLABUS

Assessment Objective	**Construct**	**Construct Description**	**Topics in the syllabus**
AO1	Force and Motion	Analysis of how force affects the state of bodies	1. Measurement and Dimensions of Physical Quantities 2. Statics 3. Linear Motion 4. Motion Under Gravity 5. Work, Energy and Power 6. Solid Friction 7. Fluid Mechanics 8. Mechanical Properties of Matter 9. Circular Motion 10. Gravitation
AO2	Energy	Exploration and application of energy forms and transformations.	1. Thermometry 2. Heat Quantities 3. Transfer of Heat 4. Behaviour of Gases 5. Thermodynamics 6. Reflection of Light 7. Refraction of Light 8. Optical Instruments 9. Simple Harmonic Motion 10. Progressive Waves 11. Stationary Waves 12. Sound Waves
AO3	Charges and Fields	Exploration of how electric and magnetic fields interact in systems and devices.	1. Electrostatics 2. Capacitors 3. Digital Electronics 4. Current Electricity 5. Magnetism in Matter 6. Magnetic Effect of an electric Current 7. Electromagnetic Induction 8. A.C Circuits
AO4	Particles	Investigation of atomic, quantum, and nuclear phenomena and their applications	1. Atomic Particles 2. Quantum Theory 3. Nuclear Processes

STRUCTURE OF THE EXAMINATION PAPERS

There will be two examination papers for Physics at Advanced Secondary Curriculum. Each examination paper will be divided into different sections with each section addressing an assessment objective.

Paper 1 (Theory Paper) will contain four sections A, B, C and D. The items will be scenario based. Section A will comprise of one item from construct four; Particles addressing assessment objective 4. Section B, C and D will comprise of two items each from construct~~s~~ one; Force and Motion, construct two; Energy and construct three; Charges and Fields addressing assessment objective 1, 2 and 3respectively.

The learner will attempt four items, including the one compulsory item from section A and one item from each of the sections B, C and D. The entire paper will take 2hours and 40 minutes.

Paper 2 (Practical Paper) will contain **two items**. Items in this paper will come from any of the four constructs. The items will assess Practical abilities of the learners in the different constructs. The items in the paper will be scenario based and the entire paper will take 2hours and 45 minutes (including time for reading, analysis of the scenario, planning and carrying out the investigation).The learner will attempt only **one item**.

Skills to be assessed:

i. Planning to select techniques, apparatus and materials;
ii. Identifying variables and controlling them;
iii. Making predictions based on prior knowledge and hypotheses;
iv. Manipulation of the apparatus and performing experiments to obtain readings/results;
v. Making and recording observations and measurements to the right accuracy of instruments and calculation;
vi. Presentation of data in an appropriate form, such as graphical and any other suitable form.
vii. Manipulating graphical data by determining gradients, intercepts or any other required points and identifying patterns in outcomes;
viii. Drawing conclusions from observations made;
ix. Assessing the suitability of procedure by identifying errors, limitations of measurements and /or experimental procedures used and suggesting methods of minimizing the errors for better results;

UGANDA ADVANCED CERTIFICATE OF EDUCATION
P510/1 SAMPLE END OF CYCLE PHYSICS THEORY EXAMINATION PAPER

Time: 2 hours and 40 minutes

Instructions:

1. Attempt **four** questions in total. Each question carries **20 marks**.
2. You must attempt **one question from each of the four sections**:
3. Section A is compulsory.
4. Use the provided data and support materials to guide your solutions.
5. Clearly show all calculations and assumptions.

SECTION A

Item 1

The display screens of digital devices use Light Emitting Diodes (LEDs), which emit high amounts of blue light. Scientific research has shown that long exposure to such light is harmful to the human eyes. An ICT student of senior five in a certain school is worried that their eyes may be at a risk due to continuous use of computers.

Support Material Provided:

- Wavelength of blue light = 450 nm
- Planck's constant, $h = 6.63\times10^{-34}$ Js,
- speed of light in vacuum, $c = 3.0\times10^{8}$ m/s
- Damage threshold for the human eye = 3.1 eV

Tasks:

(a) Use the information provided to dispel fears of the student.

(b) Explain to the student some of the symptoms of eye defects which can arise from prolonged exposure to radiations of energy beyond certain threshold.

(c) Based on principles of Physics, suggest possible protective strategies which the student can employ to safeguard his/her eyes.

SECTION B

Item 2

In a hilly district in Uganda, a landslide buried part of a school compound during the rainy season. The Ministry of Education and Sports is seeking practical proposals from physics students to help schools in the district to construct retaining structures that can prevent future landslides effects in schools in the district and other districts prone to landslides.

Support Material Provided:

Slope inclination is 45°, mass of clay soil swept by rain is 3000 kg with density 1,800 kg m^{-3}. Rain increases soil mass by 25% and Coefficient of friction, $\mu = 0.45$.

Tasks:

(a) With evidence, explain what happens when the soil gets saturated.

(b) Propose with justification, the materials that can be used to construct a realistic vertical retaining structures in the district that can be used to protect the classrooms.

Item 3

Many rural areas experience frequent motorcycle accidents due to sharp curves and poor road design. Local council authorities seek physics-based input on safe road redesign.

Support Material Provided:

Radius of curved road = 12 m, banking angle = 5°, Coefficients of friction: 0.6 (dry), 0.35 (wet). Mass of motorcycle + rider = 180 kg.

Tasks:

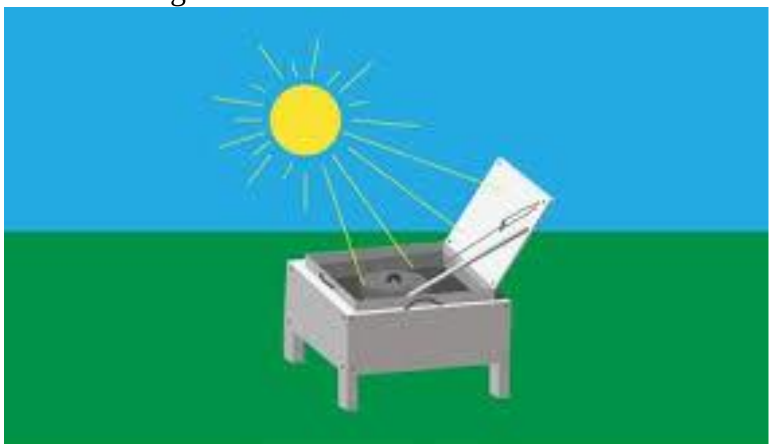
(a) With justification, advise the local council on the maximum safe speed that the riders should move at, on such roads under wet and dry conditions.

(b) Suppose that the curved road was not banked, provide convincing advice to the local council authorities, the design that would enable the motorcyclists reach their destinations earlier in both scenarios.

SECTION C

Item 4

Many rural households rely on firewood for cooking, leading to deforestation and health problems. In a bid to save our environment, the government recommended that alternative energy sources like solar be used. In order to demonstrate to the community members,a solar cooking technology with Solar irradiance of 800 W/m^2, having cooking pot area of 0.2 m^2 and efficiency of 35% is used to boil 2 kg of water from 25^oC to 100^oC. The community members wondered how this technology works and how long it would take to boil the given mass of water.



Tasks:

(a) Help the community members understand how the technology works and show to them how long it would take the water to boil.

(b) Suggest design improvements, based on thermal principles, that can cause the water to boil in a shorter time than in (a) above.

Item 5

A new rural health center is built in a valley where mobile phone signals are weak. You are part of a student team challenged to design a light-based communication system between the health center and a village watch tower located 150 meters away on a hill, in order to ease communication at the health centre. Hint: refractive indices of glass and air are 1.5 and 1.0, respectively.

Tasks:

a) Explain how communication occurs from the watch tower to the health center through the medium identified.

b) Suggest **practical improvements** that the team can implement to increase clarity or strength of the signal at the health center.

SECTION D

Item 6

Market vendors have over the years complained of destructive fire outbreaks caused by overloaded electric power extension cables. These fire outbreaks have often left market vendors devasted due to loss of property. To alleviate this situation, the chairman of the market vendors' association requested you, an A'level student of physics, to evaluate and recommend safer electrical practices.

During your evaluation, you find out that: the market is wired using an electrical cable of length 20m of resistance 0.5 $\Omega\ m^{-1}$ carrying a current of 8A for a duration of 4 hours per day. Maximum safe energy dissipation by the cable is 331.2 kJ m^{-1}.

Tasks:

(a) Analyze the information provided and suggest the cause of fire in the market.

(b) Propose practical safety improvements and awareness strategies to the vendors to avoid such occurrence.

Item 7

A farming community bordering a game reserve regularly suffers crop destruction due to wild animals entering their fields at night. To address this, they intend to develop a solar-powered electronic alarm system that uses a magnetic sensor to detect the motion of animals. They heard that such a system operates on alternating current (A.C) converted from solar energy. When activated, the system triggers a light-emitting diode (LED) to display the animal detection waveform on a Cathode Ray Oscilloscope (CRO), which has a Y-sensitivity of 20 Vcm^{-1} and the time base of 0.01 seconds, showing two full cycles of a 4 cm peak-to-peak sinusoidal waveform that generates a loud alarm.
Task:
You have been approached as a student of Physics:
a) Help the local community understand how the system works.
b) Design a diagram showing how A.C from the solar inverter can be fully rectified to power the system.

END

SCORING RUBRICS FOR PHYSICS PAPER 1

Item 1

Output / Task	Basis of Evaluation	Score				
		5	**4**	**3**	**2**	**1**
(a) Assessment likelihood of eye damage basing on energy of radiation	Correct formula, substitution, energy calculation	Formula correctly stated, constants substituted with correct SI units, energy computed accurately in J and eV,	Formula correctly stated, constants substituted with correct SI units, energy computed accurately in J and eV,	Formula correctly stated, constants substituted with correct SI units, energy computed accurately in J, fails to convert to eV	Formula correctly stated, constants substituted correctly but not in SI units, energy computed with errors in J, fails to convert to eV	Incorrect formula stated, constants substituted correctly but not in SI units, energy computed with errors in J, fails to convert to eV

	Comparison to threshold	Correct comparison to threshold and conclusion made, basing on the calculated values	Correct comparison to threshold with inappropriate conclusion made in relation to the calculation	Correct comparison to threshold and no conclusion made.	Incorrect comparison to threshold and no or inappropriate conclusion made.	No comparison to threshold and only disconnected information is provided
(b) Biological effects	Identification & explanation	At least 5 valid symptoms clearly explained.	At least 4 valid symptoms clearly explained	At least 3 valid symptoms clearly explained.	At least 2 valid symptoms clearly explained.	At least 1 valid symptom stated
(c) Protective strategies	Suggestion & justification	At least 5 valid strategies stated and clearly explained.	At least 4 valid strategies stated and clearly explained.	At least 3 valid strategies stated and clearly explained.	At least 2 valid strategies stated and clearly explained.	At least 1 valid strategy stated

Item 2

Output / Task	**Basis of Evaluation**	**Score**				
		5	**4**	**3**	**2**	**1**
a) Forces leading to landslide	Description of physical processes that take place when soil becomes saturated	States and accurately explains at least 4 valid processes	States and accurately explains at least 3 valid processes.	States and accurately explains at least 2 valid processes	States and accurately explains at least 1 valid process	States at least 1 valid process e

	Calculation of the forces that lead to land slides	Accurately calculates the forces leading to the land slide, basing on correct formula, substitution and units	Accurately calculates the forces leading to the land slide, basing on correct formula, substitution but makes minor errors on units	Calculates the forces leading to the land slide with minor errors like wrong resolution of forces and wrong units	Tries to calculate the forces leading to the land slide with errors in resolution and units	Fails to calculate the forces leading to the landslides but only tries to state some formulae or data to use
(b) Materials, their structural properties and structural adjustments to prevent soil sliding	Identification of structural properties of materials to prevent land slides	Identifies at least three materials and explains the structural properties for their suitability	Identifies two materials and explains the structural properties for their suitability	Identifies one material and explains the structural properties for its suitability	Identifies one material but the explanation for the structural properties for its suitability is incoherent	Only identifies materials without any explanation
	Description of structural adjustments or how the materials will be used	At least 5 valid adjustments stated and clearly explained	At least 4 valid adjustments stated and clearly explained	At least 3 valid adjustments stated and clearly explained	At least 2 valid adjustments stated and clearly explained	At least 1 valid adjustment stated

Item 3

Output / Task	**Basis of Evaluation**	**Score**				
		5	**4**	**3**	**2**	**1**
(a) Maximum Safe Speed Calculation	Correct formula derivation and	Correctly derives the expression for	Correctly derives the expression for	Uses the expression for maximum	Tries to use the expression for	Tries to calculate the maximum speed for

(Dry and Wet Conditions)	application; correct substitution of values	maximum speed for banked track with friction; Calculates accurately the maximum speed for both dry and wet conditions;	maximum speed for banked track with friction; Calculates the maximum speed for both dry and wet conditions with minor errors like wrong or missing units;	speed on banked track with friction to calculate the maximum speed for both dry and wet conditions;	maximum speed on banked track with friction to calculate the maximum speed for both dry and wet conditions;	both dry and wet conditions;
	Correct inferences or conclusions basing on the calculations	Provides practically safe advice to the LC that is based on the calculated values	Provides practically safe advice to the LC that is not entirely based on the calculated values	Provides practically safe advice to the LC but fails to link the calculated values to the advice	Provides inaccurate safety advice to the LC that partially links with calculated values	Provides inaccurate safety advice to the LC and is not based on the calculated values
(b) Road Design Advice (Unbanked Scenario)	Proper derivation of the condition for unbanked roads	Correctly derives the expression for maximum speed for unbanked track with friction; Calculates accurately the maximum speed for both dry and wet	Correctly derives the expression for maximum speed for unbanked track with friction; Calculates the maximum speed for both dry and wet conditions	Uses the expression for maximum speed on unbanked track with friction to calculate the maximum speed for both dry and wet conditions;	Tries to use the expression for maximum speed on unbanked track with friction to calculate the maximum speed for both dry and wet	Tries to calculate the maximum speed for both dry and wet conditions;

		conditions ;	with minor errors like wrong or missing units;		conditions ;	
	Proposal of redesign strategy based on physics principles	Provides practically proposal to the LC basing on the derived expressions and calculation.	Provides practically proposal to the LC but does not entirely link them to the derived expressions and calculation.	Provides some safety advice to the LC but fails to link them to the calculations	Provides inaccurate safety advice to the LC but attempts to link the advice to the calculations	Provides inaccurate safety advice to the LC and is not connected to the expressions and calculated values

Item 4

Output / Task	Basis of Evaluation	Score				
		5	**4**	**3**	**2**	**1**
a) Working of a solar heater	Steps in calculation of usable power and how long it takes to boil water	Correctly calculates energy required; determines usable power and computes boiling time accurately with correct units.	Correctly calculates energy required; determines usable power and computes boiling time accurately with correct units.	Calculates energy required; determines usable power and computes boiling time.	Calculates energy required; determines usable power and computes boiling time	Tries to calculate energy required; and boiling time
	Understanding of solar	Identifies and explains at least four	Identifies and explains any three	Identifies and explains any two	Identifies and explains any one	Mentions at least one principle

	heating principles	solar heating principles	solar heating principles	solar heating principles	solar heating principles	but does not explain
b) Design Improvements	Suggests feasible improvements based on thermal principles	Proposes at least five practical, physics-based improvements with clear justification	Proposes at least four practical, physics-based improvements with clear justification	Proposes at least three practical, physics-based improvements with clear justification	Proposes at least two practical, physics-based improvements with clear some justification	Proposes any practical, physics-based improvement.

Item 5

Output / Task	**Basis of Evaluation**	**Score**				
		5	**4**	**3**	**2**	**1**
(a) Explanation of Communication	Understanding of light propagation through the medium	Provides a clear, accurate explanation with illustrations of optical communication processes, through free space and optical fibres with relevant mathematical backup like calculation of critical angle	Provides a clear, accurate explanation of optical communication processes, through free space and optical fibres with relevant mathematical backup like calculation of critical angle.	Provides a clear, accurate explanation of optical communication processes, through free space and optical fibres with limited mathematical backup.	Provides an explanation of optical communication processes, through either free space and/or optical fibres with no mathematical back up	States optical communication processes, through either free space and/or optical fibres in an incoherent way
(b) Practical Improvement Suggestions	Feasibility and	Proposes at least five realistic,	Proposes any four realistic,	Proposes any three realistic,	Proposes any two realistic,	Proposes any one justificatio

	physics reasoning	physics-based with clear justification for each of free space and optical fibres.	physics-based with clear justificatio n for each of free space and optical fibres.	physics-based with clear justification for each of free space and optical fibres.	physics-based with justification for each of free space and/or optical fibres.	n for each of free space and/or optical fibres

Item 6

Output / Task	Basis of Evaluatio n	Score				
		5	**4**	**3**	**2**	**1**
(a) Analysis of Fire Cause	Correct calculatio n of energy dissipatio n	Correctly calculates energy dissipation and compares with the safe limits;	Correctly calculates energy dissipation with minor errors like missing units and compares with the safe limits;	Calculates energy dissipation with major errors like missing units and incorrect substitution, compares with the safe limits;	Tries to calculate energy dissipation incoherentl y, and compares with the safe limits	Attempts to calculate the energy dissipatio n but can identify just a few relevant formulae
	Correct identificat ion of electrical overload and reasoning basing on calculated data	Clearly explains the cause of fire in relation to energy dissipation Correctly applies obtained results and energy concepts to make valid conclusion.	Clearly explains the cause of fire in relation to calculated values of energy Correctly applies obtained results and energy concepts to make valid conclusion	Explains the cause of fire; Applies obtained results and energy concepts to make a conclusion.	Explains the cause of fire without referring to the calculation of energy dissipation	Explains the cause of fire without referring to any calculatio n

(b) Safety Improvement Suggestions	Practical and feasible measures	Proposes at least five realistic measures with clear physics justification	Proposes at least four realistic measures with clear physics justification	Proposes at least three realistic measures with justification.	Proposes at least two realistic measures with some justification	Proposes some measures without any or with false justification.

Item 7

Output / Task	**Basis of Evaluation**	**Score**				
		5	**4**	**3**	**2**	**1**
(a) Explanation of System Operation	Understanding of solar-powered magnetic detection and CRO display	Clear, accurate explanation of how solar energy is converted to electrical A.C., how the magnetic sensor detects motion, and how the LED/CRO displays the waveform; Correctly explains induced voltage, current flow, and signal display with well labelled diagrams;	Clear, accurate explanation of how solar energy is converted to electrical A.C., how the magnetic sensor detects motion, and how the LED/CRO displays the waveform; Correctly explains induced voltage, current flow, and signal display with partially labeled diagrams.	Explains how solar energy is converted to electrical A.C., how the magnetic sensor detects motion, and how the LED/CRO displays the waveform; Explains induced voltage, current flow, and signal display without fully labeled diagrams.	Tries to explain how solar energy is converted to electrical A.C.; Tries to explain how the magnetic sensor detects motion, and how the LED/CRO displays the waveform without fully labeled diagrams.	States how solar energy is converted to electrical A.C.; States how the magnetic sensor detects motion, and how the LED/CRO displays the waveform

(b) Rectification Diagram	Completeness, accuracy, and clarity of schematic full wave rectification	Draws a completely labeled diagram showing full-wave rectification;	Draws a labeled diagram showing full-wave rectification;	Draws a labeled diagram showing full-wave rectification;	Draws a labeled diagram showing full-wave rectification;	Draws a diagram showing full-wave rectification;
	Correct explanation of the components	Correctly explains the function of at least 8 labelled components to ensure full wave rectification.	Correctly explains the function of at least 6 labelled components to ensure full wave rectification	Correctly explains the function of at least 4 labelled components to ensure full wave rectification	Explains the function of at least 2 labelled components to ensure full wave rectification	Tries to explains the function of any labeled component

END

SAMPLE END OF CYCLE ADVANCED LEVEL P510/2 PHYSICS PRACTICAL EXAMINATION ITEMS

Time Allowed: 2 hours

Instructions to Candidates:

Attempt **only one** item

1. For each task:
   a) Identify the problem or relationship and hypothesis to investigate.
   b) Design and carry out a valid experiment.
   c) Record and present your data clearly.
   d) Analyze your results and draw reasoned conclusions.
   e) Reflect on accuracy of the results and suggest improvements.
2. Support materials for each task are provided.

Item 1

A local boda-boda (motorcycle) garage is interested in improving the fuel efficiency and safety of motorcycles by reducing unnecessary load and friction. Mechanics often claim that heavier loads make motorcycles harder to accelerate, but they cannot quantify this effect. The garage consults physics students to investigate the above phenomenon using a trolley model.

Task:

As a Physics student, you are required to conduct an experiment to reassure the mechanics about their claim.

Item 2

An electrician observes that a wire in a household circuit heats up significantly when current flows through it. He suspects that this wire may have an unusual electric behaviour and fears that it might cause fire and burn the house. He therefore wishes to replace the wire with another one, but has two alternatives to choose from.

Task

a) As a physics student, carry out an investigation to help the electrician to select the right wire for replacing the wire that heats up.

b) At a chosen value of V, compare the power generated from each conductor and advise the electrician accordingly.

END

ADVANCE INFORMATION FOR P510/2 PHYSICS PRACTICAL

Item 1

- Trolley
- A smooth runway
- Light pulley
- Slotted masses and a hunger(50-500 g)
- A string
- A meter rule
- A retort stand and a clamp
- A stop clock/watch

Item 2

- DC power supply (adjustable voltage)
- Ammeter and voltmeter (digital or analogue)
- Conducting wires of different materials (e.g. copper, nichrome, constantan)
- Connecting wires
- Resistors of known values
- Switch
- Rheostat (variable resistor)
- Micrometer screw gauge or vernier caliper (to measure wire thickness)
- Stopwatch (for timed heating trials)

SCORING RUBRIC FOR ADVANCED LEVEL PRACTICAL PHYSICS EXAMINATION

Item 1

Output / Task	Basis of Evaluation	Score				
		5	4	3	2	1
Aim	Clarity and relevance of the aim	Clear, concise, and fully relevant aim linked to investigation relating mass to acceleration.	Mostly clear and relevant aim linked to investigation relating mass to acceleration	Relevant aim linked to investigation relating mass to acceleration	Aim linked to investigation relating mass to acceleration	Aim has little or no link to the investigation
Hypothesis	Appropriateness and testability	Well-stated, testable hypothesis; clearly predicts the relationship between mass and acceleration.	Relevant and testable hypothesis; clearly predicts the relationship between mass and acceleration.	Partially stated hypothesis to predict the relationship between mass and acceleration	Unclear hypothesis stated that does not lead to prediction of the relationship between mass and acceleration	Irrelevant or no hypothesis stated
Independent Variable	Correct identification	Correctly identifies and accurately explains the independent variables	Correctly identifies and partially explains the independent variables	Correctly identifies the independent variables without explanation	Partial identification of independent variables	Variables not distinguished
Dependent Variable	Correct identification	Correctly identifies and accurately explains the	Correctly identifies and partially explains the	Correctly identifies the dependent variables without explanation	Partial identification of dependent variables	Variables not distinguished

		dependent variables.	dependent variables.			
Controlled Variables	Completeness and correctness	Correctly identifies and accurately explains the controlled variables	Correctly identifies and partially explains the controlled variables	Correctly identifies the controlled variables without explanation	Partial identification of controlled variables	Controlled Variables not distinguished
Procedure	Clarity, completeness, logical sequence	Clear, concise, replicable and fully relevant step-by-step procedure establishing the relationship between mass and acceleration	Clear, replicable and fully relevant step-by-step procedure establishing the relationship between mass and acceleration.	Replicable and relevant step-by-step procedure establishing the relationship between mass and acceleration	Partially relevant procedure establishing the relationship between mass and acceleration	Totally illogical procedure which does not lead to establishing the relationship between mass and acceleration
Data Presentation	Organization, tables, units	Data presented clearly in correct columnar table with units and labels indicated; measurements recorded to the accuracy of the instrument used.	Data presented clearly in correct columnar table with some errors in units and labels; measurements recorded to the accuracy of the instrument used.	Data presented in columnar table with some errors in units and labels; measurements recorded to the accuracy of the instrument used.	Data presented in an incomplete table with errors in units and labels; measurements recorded inconsistently.	Totally illogical data with no trends presented

Errors and Precautions	Identification and explanation	Accurately identifies errors; accurately explains their sources and proposes appropriate precautions	Accurately identifies errors; partially explain their sources and proposes appropriate precaution s.	Identifies errors; and proposes appropriate precautions	Identifies some errors and suggests inappropriate precautions	Irrelevant sources of errors and precautions stated
Data Analysis (Graphical)	Accuracy, interpretation	Title clearly stated; axes drawn and labeled; Appropriate scale; points correctly plotted; trend analyzed accurately	Title clearly stated; axes drawn and labeled with missing units; Appropriate scale; points correctly plotted; trend analyzed accurately.	Title stated; axes drawn and labeled with missing units; inappropriate scale; points plotted; trend analyzed.	Major errors in either axes labels, finding the scale, plotting, or analyzing trends.	Irrelevant analysis of data, like plotting wrong graph
Conclusion /Recommendation	Justification using physics principles	Accurate, logical conclusion, fully justifying the investigation carried out; recommendations appropriate basing on the results.	Clear, logical conclusion, fully justifying the investigation carried out; recommendations basing on the results.	Makes a conclusion partly justifying the investigation carried out with partial recommendation.	Makes a conclusion that hardly justifies the investigation carried out with some recommendation.	Irrelevant conclusion or incorrect recommendation.

Item 2

Output / Task	Basis of Evaluation	Score				
		5	**4**	**3**	**2**	**1**
Aim	Clarity and relevance of the aim	Clear, concise, and fully relevant aim linked to investigation relating I to V.	Mostly clear and relevant aim linked to investigation relating I to V.	Relevant aim linked to investigation relating I to V	Aim linked to investigation relating I to V.	Aim has little or no link to the investigation
Hypothesis	Appropriateness and testability	Well-stated, testable hypothesis; clearly predicts the relationship between I to V.	Relevant and testable hypothesis; clearly predicts the relationship between I to V.	Partially stated hypothesis to predict the relationship between I to V.	Unclear hypothesis stated that does not lead to prediction of the relationship between I to V.	Irrelevant or no hypothes is stated
Independent Variable	Correct identification	Correctly identifies and accurately explains the independent variables.	Correctly identifies and partially explains the independent variables.	Correctly identifies the independent variables without explanation	Partial identification of independent variables	Variables not distinguished
Dependent Variable	Correct identification	Correctly identifies and accurately explains the dependent variables.	Correctly identifies and partially explains the dependent variables	Correctly identifies the dependent variables without explanation	Partial identification of dependent variables	Variables not distinguished
Controlled Variables	Completeness and correctness	Correctly identifies and accurately explains the controlled variables.	Correctly identifies and partially explains the controlled variables.	Correctly identifies the controlled variables without explanation	Partial identification of controlled variables	Controlled Variables not distinguished

Procedure	Clarity, completeness, logical sequence	Clear, concise, replicable and fully relevant step-by-step procedure establishing the relationship between I to V.	Clear, replicable and fully relevant step-by-step procedure establishing the relationship between I to V.	Replicable and relevant step-by-step procedure establishing the relationship between I to V.	Partially relevant procedure establishing the relationship between I to V.	Totally illogical procedure which does not lead to establishing the relationship between I to V.
Data Presentation	Organization, tables, units	Data presented clearly in correct columnar table with units and labels indicated; measurements recorded to the accuracy of the instrument used.	Data presented clearly in correct columnar table with some errors in units and labels; measurements recorded to the accuracy of the instrument used.	Data presented in columnar table with some errors in units and labels; measurements recorded to the accuracy of the instrument used.	Data presented in an incomplete table with errors in units and labels; measurements recorded inconsistently.	Totally illogical data presented, with no trends
Errors and Precautions	Identification and explanation	Accurately identifies errors; accurately explains their sources and proposes appropriate precautions	Accurately identifies errors; partially explain their sources and proposes appropriate precautions	Identifies errors; and proposes appropriate precautions	Identifies some errors and suggests inappropriate precautions.	Irrelevant errors and precautions stated

Data Analysis (Graphical)	Accuracy, interpretation	Title clearly stated; axes drawn and labeled; Appropriate scale; points correctly plotted; trend analyzed accurately.	Title clearly stated; axes drawn and labeled with missing units; Appropriate scale; points correctly plotted; trend analyzed accurately.	Title stated; axes drawn and labeled with missing units; inappropriate scale; points plotted; trend analyzed	Major errors in either axes labels, finding the scale, plotting, or analyzing trends	Irrelevant analysis of data, like plotting wrong graph
Conclusion/Recommendation	Justification using physics principles	Accurate, logical conclusion, fully justifying the investigation carried out; recommendations appropriate basing on the results.	Clear, logical conclusion, fully justifying the investigation carried out; recommendations basing on the results	Makes a conclusion partly justifying the investigation carried out with partial recommendation	Makes a conclusion that hardly justifies the investigation carried out with some recommendation	Irrelevant conclusion or incorrect recommendation

OVERALL GRADE DESCRIPTORS

Construct 1: Forces and Motion

The learner investigates how forces affect motion and structures, and uses this knowledge to design safe, efficient, and practical solutions in everyday situations.

Grade A (Exceptional)	Investigates and analyses complex force–motion interactions; accurately interprets data to make logical inferences and designs innovative, efficient, and safe mechanical or structural solutions for real-life problems.
Grade B (Outstanding)	Effectively investigates how forces affect motion and structures; applies correct concepts to develop appropriate innovations, efficient solutions with minimal errors.
Grade C (Satisfactory)	Investigates straightforward interactions of forces and motion; applies known methods to suggest workable solutions with some innovations.

Grade D (Basic)	Identifies simple force-motion situations with limited interpretation; shows partial understanding and limited ability to relate knowledge to practical or experimental examples.
Grade E (Elementary)	Shows limited identification or application of force and motion interaction in real life problems.

Construct 2: Energy Transfer and Transformation

The learner investigates how energy is transferred and transformed through heat, light, sound, and waves, and applies this understanding to design, use, and explain devices and systems in daily life, industry, and technology.

Grade A (Exceptional)	Analyses energy data in a variety of situations to make informed decisions and innovates or models complex systems of energy transfer with precision; explains, evaluates, and optimizes technological devices for efficiency.
Grade B (Outstanding)	Investigates and explains multiple forms of energy transformations; applies principles effectively to innovatively design or improve devices with sound reasoning.
Grade C (Satisfactory)	Demonstrates understanding of common energy conversions; correctly designs with minimal errors, describes or operates simple devices.
Grade D (Basic)	Recognises simple forms of energy transfer; states incomplete or partly correct innovative designs and explanations of simple device operation.
Grade E (Elementary)	Shows minimal awareness of energy concepts; unable to design, describe or demonstrate energy transfer in simple devices.

Construct 3: Electricity and Magnetism

The learner explores how electric and magnetic fields interact in systems and devices, and uses this understanding to explain and model technologies for power transmission, energy storage, and electronic control in daily life and industry.

Grade A (Exceptional)	Demonstrates comprehensive understanding of electric and magnetic field interactions; innovatively models, analyses, and predicts behaviour of complex systems and technologies.
Grade B (Outstanding)	Accurately applies principles of electricity and magnetism to explain and innovatively model common devices (motors, transformers, generators) with minimal errors.
Grade C (Satisfactory)	Understands core concepts of electricity and magnetism; can innovatively design, describe and test basic electrical or magnetic systems.
Grade D (Basic)	Shows partial understanding of electricity and magnetism principles; struggles to connect theory to practical devices or applications.
Grade E (Elementary)	Has very little understanding of electricity or magnetism principles or their interactions; cannot describe or design basic electrical or magnetic systems.

Construct 4: Atomic, Quantum, and Nuclear Phenomena
The learner investigates atomic, quantum, and nuclear phenomena, and uses this understanding to explain how modern technologies work and to assess the benefits and risks of their applications.

Grade A (Exceptional)	Integrates advanced understanding of atomic, quantum, and nuclear physics to explain and critically evaluate modern technologies and their social or environmental impacts.
Grade B (Outstanding)	Accurately explains and applies key principles of atomic and nuclear physics to interpret technological applications, showing analytical insight.
Grade C (Satisfactory)	Describes essential atomic and nuclear concepts and relates them to simple real-world technologies (e.g., X-rays, radioactivity) with minimal errors.
Grade D (Basic)	Recognises a few atomic and nuclear principles and gives incomplete or inaccurate explanations of phenomena or applications.
Grade E (Elementary)	Shows limited understanding of atomic and nuclear concepts; partially relates them to modern technology applications.

SECTION B ITEMS (FORCE AND MOTION)

Item 1: Motion under Gravity, Work, Energy and Power, Mechanical Properties of Matter

A thrill-seeker of mass 60.0 kg jumps from a bridge with a bungee cord of natural length 15.0 m. The cord behaves like a spring with spring constant $k=200Nm^{-1}$. Assume no air resistance and that the cord obeys Hooke's Law.

Tasks:

a) Calculate the speed of the jumper just as the cord begins to stretch.
b) Determine the maximum extension of the cord using the principle of conservation of energy.
c) Explain why the jumper does not return to the original drop point after bouncing back up.

Item 2: Fluid Mechanics (Pascal's Principle), Statics (Equilibrium), Work, Energy and Power

A garage uses a hydraulic lift to raise cars. The lift consists of two connected cylinders: a small input piston of cross-sectional area 0.01 m^2 and a large output piston of area 0.50 m^2. A car of mass 1500 kg rests on the large piston.

Tasks:

a) Calculate the force that must be applied to the small piston to just lift the car.
b) If the small piston moves down by 0.50 m, how far does the car rise?
c) Calculate the work done on the small piston and the work done on the car. What principle does this demonstrate?

Item 3: Skidding to a Stop

Topics: Linear Motion, Solid Friction, Work, Energy and Power

A driver of a car of mass 1200 kg is traveling at $20ms^{-1}$ on a horizontal road. The coefficient of kinetic friction between the tires and the road is $\mu_k=0.60$. The driver slams on the brakes, locking all four wheels.

Tasks:

a) Calculate the frictional force that stops the car.
b) Determine the deceleration of the car.
c) Calculate the minimum stopping distance.
d) Using the work-energy theorem, verify your answer to (c).

Item 4: Conical Pendulum in a Carnival Ride

Topics: Circular Motion, Statics (Resolution of Forces), Measurement and Dimensions

A "Gravitron" ride consists of a mass, m=5.0kg attached to a string of length, L=2.0m, swinging in a horizontal circle of radius, r=1.2 m, forming a conical pendulum.

Tasks:

a) Draw a free-body diagram showing the forces acting on the mass.
b) Calculate the tension in the string.
c) Determine the speed of the mass and the period of rotation.

Item 5: Satellite Orbit and Gravitational Field

Topics: Gravitation, Circular Motion, Measurement and Dimensions (Units)

A remote sensing satellite of mass 500 kg orbits Earth at an altitude of 400 km above the surface. (Earth mass M_E=5.98×10^{24} kg, Earth radius R_E=6.37×106 m, G=6.67×10^{-11} Nm^2kg^{-2}).

Tasks:

a) State Newton's Law of Universal Gravitation and verify its dimensional homogeneity.
b) Calculate the orbital speed of the satellite.
c) Calculate the gravitational potential energy of the satellite in orbit.
d) Explain why the astronaut inside the satellite experiences weightlessness.

Item 6: Measuring Young's Modulus of a Wire

Topics: Mechanical Properties of Matter, Measurement and Dimensions (Uncertainty), Statics

A student investigates the Young's modulus of a steel wire of length 2.500 m and diameter 0.50 mm. She hangs masses from the wire and measures the extension using a micrometer. The following data is recorded: Load = 10.0 kg, Extension = 1.25 mm.

Tasks:

a) Calculate the stress and strain in the wire.
b) Calculate Young's modulus for steel.
c) The micrometer used has an uncertainty of ±0.01 mm±0.01mm. Calculate the percentage uncertainty in the diameter and the cross-sectional area.
d) State two assumptions made when calculating Young's modulus from this experiment.

Item 7: Terminal Velocity of a Raindrop

Topics: Fluid Mechanics (Viscosity/Drag), Motion under Gravity, Linear Motion

A spherical raindrop of radius r=1.5mm falls through air. The density of water is 1000 kgm^{-3}, density of air is 1.2 kgm^{-3}, and the drag coefficient, C=0.5. The drag force is given by $F=12C\rho_{air}Av^2$, where A is the cross-sectional area.

Tasks:

a) Calculate the volume and weight of the raindrop.
b) Determine the terminal velocity of the raindrop.
c) Explain why the raindrop reaches terminal velocity and sketch a velocity-time graph for its fall.

Item 8: Leaning Ladder Problem

Topics: Statics (Equilibrium, Torque), Solid Friction, Measurement and Dimensions

A uniform ladder of length, L=5.0 m and mass, M=15kg rests against a smooth vertical wall. The coefficient of static friction between the ladder and the rough horizontal ground is μ_s=0.40. A person of mass, m=70kg stands on the ladder at a distance x from the bottom.

Tasks:

a) Draw a diagram showing all forces acting on the ladder. (2 marks)
b) Determine the minimum angle, θ to the horizontal such that the ladder does not slip when the person stands at the very top (x=L). (5 marks)
c) State what happens to the friction force as the person climbs higher. (3 marks)

Item 9: Projectile Motion on a Planet

Topics: Motion under Gravity, Linear Motion (Projectiles), Gravitation

An astronaut on the Moon hits a golf ball with an initial speed of 40 m/s at an angle of 30° above the horizontal. (Moon's gravity, g_m=1.6 ms^{-2}, compare to Earth's g_e=9.8 ms^{-2}).

Tasks:

a) Calculate the time of flight of the ball.

b) Determine the horizontal range.

c) Calculate the maximum height reached.

d) Compare the range on the Moon with that on Earth for the same launch conditions, explaining the difference.

Item 10: Roller Coaster Loop

Topics: Circular Motion, Work, Energy and Power, Mechanical Properties of Matter

A roller coaster car of mass 800 kg starts from rest at point A, a height, H=30 m above the ground. It then goes around a vertical circular loop of radius R=8.0 m. Assume no friction.

Tasks:

a) Calculate the speed of the car at the top of the loop (point B). (3 marks)

b) Determine the centripetal acceleration at the top of the loop and state whether the car remains on the track. (3 marks)

c) Calculate the normal force exerted by the track on the car at the bottom of the loop. (4 marks)

Item 11: Road Construction Survey (Measurement, Statics, Friction)

A civil engineering team is surveying a sloped road. A wooden beam of length 5.0 m and mass 20 kg rests on a rough incline at 30°. The coefficient of friction between the beam and ground is 0.4. A student measures angles using a clinometer with ±1° uncertainty.

Tasks:

a) State the SI base units of force and derive its dimensions.

b) Calculate the component of the beam's weight parallel to the slope.

c) Determine whether the beam will remain at rest.

d) Estimate the percentage uncertainty in the angle measurement.

e) Explain how measurement errors affect safety in construction.

Item 12: Motorcycle Motion Analysis (Linear Motion & Friction)

A motorcycle accelerates uniformly from rest to 20 m/s in 5 s on a horizontal road, then brakes to rest over 40 m.

Tasks:

a) Determine the acceleration during the first phase.

b) Calculate the braking deceleration.

c) Determine the coefficient of friction between tyres and road.

d) Sketch a velocity-time graph for the motion.

e) Explain how friction improves road safety.

Item 13: Waterfall Power Generation (Energy, Gravity, Fluids)

Water falls from a height of 25 m at a rate of 200 kg/s in a small hydroelectric plant.

Tasks:

a) Calculate the gravitational potential energy lost per second.
b) Determine the power generated (assume 80% efficiency).
c) Explain the role of gravity in energy conversion.
d) Describe how fluid pressure varies with depth in the reservoir.
e) Suggest two ways to improve efficiency.

Item 14: Satellite Motion (Circular Motion & Gravitation)

A satellite orbits Earth at a height where gravitational acceleration is 8.7 m/s^2.

Tasks:

a) Explain why the satellite remains in orbit.
b) Derive the expression for centripetal force. (3 ma
c) Calculate the orbital speed of the satellite. (4 marks)
d) Explain the relationship between gravitation and circular motion. (3 marks)
e) State one application of satellites in Uganda. (2 marks)

Item 15: Bridge Material Testing (Elasticity & Work-Energy)

A steel wire of length 2 m and cross-sectional area $1.0\times10^{-6} m^2$ stretches by 2 mm under a load of 200 N.

Tasks:

a) Define stress and strain.
b) Calculate the Young's modulus of the wire.
c) Determine the elastic potential energy stored.
d) Explain why materials must have high elasticity for bridges.
e) State one limitation of Hooke's law.

Item 16: Rocket Launch Safety Inspection

A small test rocket of mass 200 kg is launched vertically upward. Engineers need to verify its structural limits and motion parameters.

Tasks:

(a) The rocket engine provides a constant thrust of 4000 N for the first 8 seconds. Calculate the net force and the acceleration during this phase.
(b) Determine the velocity and height of the rocket at engine cut-off (t = 8 s).
(c) After cut-off, the rocket coasts upward under gravity only. Find the maximum height reached above launch pad.
(d) The rocket's support strut has a cross-sectional area of 4×10^{-4} m^2 and a Young's modulus of 2.0×10^{11} Pa. During launch, the strut experiences a tensile stress equal to the thrust divided by total cross-sectional area (four struts). Calculate the strain in each strut.

Item 17: Hydroelectric Dam Analysis

A hydroelectric dam stores water at a height of 80 m above the turbines. The flow rate of water is 500 kg/s. The water falls through a pipe of varying diameter.

Task:

(a) Calculate the maximum electrical power available if the turbine-generator system is 75% efficient. (g = 9.8 m/s^2)

(b) At a point where the pipe diameter is 0.25 m, the water speed is 4 m/s. Find the diameter where the speed doubles, assuming incompressible flow.
(c) Using dimensional analysis, show that the power P developed by a water turbine depends on density ρ, flow rate Q, and head h as $P \propto \rho g Q h$.

Item 18: Car on Inclined Curve

A car of mass 1500 kg travels around a curved road banked at 25° to the horizontal. The radius of curvature is 120 m. The coefficient of static friction between tires and road is 0.35.

Tasks:

(a) Calculate the maximum speed the car can have without relying on friction.
(b) If the car travels at 28 m/s, find the minimum coefficient of friction needed to prevent skidding.
(c) Explain what happens to the normal reaction force as speed increases beyond the design speed.

Item 19: Falling Object with Viscous Drag

A small steel ball of radius 2.0 mm and density 7800 kg/m³ is released from rest in a tall column of glycerin (density 1260 kg/m³, viscosity 1.5 Pa·s).

Tasks:

(a) Show that the ball quickly reaches terminal velocity and calculate this terminal velocity using Stokes' law: $F=6\pi\eta rv$.
(b) Determine the time taken to reach 95% of terminal velocity if the characteristic time $\tau = \frac{m}{6\pi\eta r}$.
(c) Verify dimensional consistency of Stokes' law.

Item 20: Satellite Energy and Gravitation

A weather satellite of mass 800 kg orbits Earth at an altitude of 900 km. Earth's radius = 6400 km, mass = 6.0×10^{24} kg, $G = 6.67\times10^{-11}$ N·m²/kg².

Tasks:

(a) Calculate the orbital speed and period of the satellite. (4 marks)
(b) Determine the total mechanical energy of the satellite in orbit. (3 marks)
(c) If the satellite is to be moved to a higher orbit, explain whether work must be done on or by the satellite, and what happens to its kinetic and potential energies. (3 marks)

Item 21: Parachutist Descent Scenario

A parachutist of mass 80 kg jumps from a height of 400 m. Initially in free fall, they reach terminal velocity after deploying the parachute. Air resistance provides an upward force proportional to speed squared. On landing, they slide 10 m horizontally on rough ground with coefficient of friction $\mu=0.4$, decelerating uniformly from 5 m/s to rest.

Tasks:

a) Calculate the parachutist's speed just before parachute deployment, ignoring air resistance initially.
b) Explain using forces why terminal velocity occurs, and derive its magnitude assuming drag

equals weight at terminal speed.
c) Find the frictional force and work done by friction during sliding.

Item 22: Water Tank Fluid System

A cylindrical water tank (density, $\rho=1000\ kgm^{-3}$) of height 5 m and radius 2 m is mounted on supports. It exerts pressure on the base and walls. A small hole at depth h below the surface causes water to jet out horizontally with speed given by Torricelli's theorem. The tank's steel base (thickness 0.01 m, $Y=2\times10^{11}Pa$) compresses slightly under the weight.

Tasks:

a) Calculate the pressure at the tank base due to the water column.
b) For h = 4 m, find the horizontal speed of efflux and horizontal distance traveled by water before hitting ground 1 m below hole.
c) Estimate the compressive strain in the base from the water's weight.

Item 23: Rollercoaster Loop

A rollercoaster car of mass 500 kg enters a vertical loop of radius 10 m at 20 m/s at the bottom. Friction is negligible. Riders experience normal force from the seat, and the track provides centripetal force.

Tasks:

a) Calculate the speed at the top of the loop using energy conservation.
b) Find the centripetal acceleration at the top and the minimum speed required there for contact with the track.
c) Determine the apparent weight of a 70 kg rider at the top.

Item 24: Satellite Launch

A satellite of mass 1000 kg is launched into orbit at height 300 km above Earth (radius $R_e=6.37\times10^6$ m, $g=9.8ms^{-2}$). It requires circular orbit speed, with gravitational force providing centripetal acceleration. Launch involves vertical motion under gravity from a rocket.

Tasks:

a) Calculate gravitational acceleration at orbital height.
b) Find orbital speed and period for circular motion.
c) If launched vertically from rest at ground with initial kinetic energy, estimate minimum energy needed to reach 300 km (ignore drag).

Item 25: The Logistics Warehouse Challenge

Topics Covered: Statics, Solid Friction, Work, Energy, and Power.

A logistics company in Kampala uses a motorized ramp to load crates into trucks. A crate of mass 80 kg is placed on a ramp inclined at 20° to the horizontal. The coefficient of static friction between the crate and the ramp is 0.4, and the coefficient of kinetic friction is 0.3.

Task:

a) Analyze whether the crate will remain at rest on the ramp without the motor's assistance.
b) If the motor pulls the crate 5 meters up the ramp at a constant speed of $2ms^{-1}$, calculate the total work done by the motor.
c) Determine the minimum power rating the motor requires to complete this task.

Item 26: The "Pearl of Africa" Satellite Orbit

Topics Covered: Gravitation, Circular Motion, Measurement and Dimensions.

Uganda's Space Agency is designing a low-earth orbit satellite for environmental monitoring. The lead engineer proposes a formula for the orbital period T as $T = k.R^x.G^y.M^z$, where R is the orbital radius, G is the gravitational constant, M is the mass of the Earth, and k is a dimensionless constant.

Task:

a) Use Dimensional Analysis to find the values of x, y, and z.
b) The satellite is intended to orbit at an altitude where the acceleration due to gravity is 80% of its value at the Earth's surface. Calculate the required orbital speed for this satellite to maintain a stable circular path.
c) Explain how the satellite's weightlessness in orbit is a result of circular motion principles rather than a lack of gravity.

Item 27: The Deep-Lake Research Probe

Topics Covered: Fluid Mechanics, Mechanical Properties of Matter, Statics.

A spherical research probe is lowered into Lake Bunyonyi to study the lake bed. The probe is suspended by a steel cable of length 40 m and diameter 12 mm. The probe has a volume of $0.5m^3$ and a mass of 600 kg. (Density of water = $1000kgm^{-3}$; Young's Modulus of steel = 2.0×10^{11}Pa).

While the probe is fully submerged and stationary, calculate the upthrust acting on it and the resulting tension in the cable.

Task:

a) Determine the elastic extension of the steel cable caused by the submerged probe.
b) If the water pressure at the bottom is 4.0×10^5Pa, estimate the fractional change in the volume of the probe if its Bulk Modulus is 1.5×10^{10}Pa.

Item 28: The Solar-Powered Water Pump

Topics Covered: Linear Motion, Motion under Gravity, Work, Energy, and Power.

A farm in a hilly region uses a solar pump to lift water from a valley stream to a reservoir on a hill 30 meters higher. The water is ejected from a pipe at the reservoir at a speed of 4m/s. The pump moves 2kg of water every second.

Task:

a) Calculate the total kinetic energy and potential energy gained by the water per second.
b) The water falls from the reservoir pipe back into a trough. Determine the velocity of the water just before it hits the trough, which is 2 meters below the pipe outlet.
c) If the solar panels provide 1.2 kW of power to the pump, calculate the efficiency of the pumping system.

Item 29: The High-Speed Railway Prototype

Topics Covered: Linear Motion, Solid Friction, Mechanical Properties of Matter.

Engineers are testing a prototype rail carriage on a horizontal track. The carriage of mass 2,000kg is accelerated from rest to 30m/s over a distance of 200 m. To stop the carriage, high-friction emergency brake pads are applied, creating a constant frictional force of 6,000 N.

Task:

a) Calculate the uniform acceleration of the carriage during the start-up phase.
b) During braking, the steel brake rod (length 0.8 m) experiences a compressive stress. If the rod must not compress by more than 0.5 mm, determine the minimum cross-sectional area required for the rod. (Young's Modulus for the rod = 2.1×1011Pa).
c) Evaluate the distance the carriage travels while braking to a complete stop.

Item 30: Measuring Acceleration of a Falling Object

Topics: Measurement & Dimensions, Motion under Gravity, Linear Motion

A student drops a small steel ball from rest at a measured height of 2.0 m above a sand surface. Using a stopwatch, the student records the time taken for the ball to hit the sand as 0.64 s.

Tasks:

(a) Calculate the average acceleration of the ball during its fall.

(b) Determine the experimental value of g and compare it with the accepted value of 9.8 ms^{-2}.

(c) Discuss two possible sources of error in the measurement.

(d) Explain how dimensions of physical quantities can be used to check the correctness of the formula $s=ut+\frac{1}{2}at^2$.

Item 31: Static Equilibrium of a Beam

Topics: Statics, Solid Friction, Mechanical Properties of Matter

A uniform beam of length 4 m and weight 200 N is supported horizontally at its ends. A student of weight 600 N stands 1.5 m from the left end.

Tasks:

(a) Draw a labeled diagram showing all forces acting on the beam.

(b) Calculate the reaction forces at the supports.

(c) If the beam is replaced with one made of lighter material but same length, explain how the reactions change.

(d) Discuss the role of solid friction if the supports are not fixed.

Item 32: Circular Motion of a Car

Topics: Circular Motion, Solid Friction, Work, Energy & Power

A car of mass 1200 kg moves at constant speed of 20 m/s around a flat circular track of radius 50 m.

Tasks:

(a) Calculate the centripetal force acting on the car.

(b) If the coefficient of friction between the tires and the road is 0.4, determine whether the car can safely negotiate the curve.

(c) Explain how banking the road would reduce reliance on friction.

(d) Using energy principles, discuss what happens if the car exits the curve onto a straight road.

Item 33: Fluid Mechanics and Pressure

Topics: Fluid Mechanics, Mechanical Properties of Matter, Work, Energy & Power

A cylindrical tank of radius 1.2 m is filled with water to a height of 3.0 m.

Tasks:

(a) Calculate the pressure at the base of the tank.

(b) Determine the force exerted on the base.

(c) If a small hole is opened at the bottom, calculate the speed of efflux of water.

(d) Explain how the mechanical properties of matter (density and compressibility) affect the behavior of water in the tank.

Item 34: Gravitational Potential Energy in Satellites

Topics: Gravitation, Circular Motion, Work, Energy & Power, Measurement & Dimensions

A satellite of mass 500 kg is placed in a circular orbit of radius 7.0×106 m around Earth (Earth's mass = 6.0×10^{24} kg, G=6.67×10^{-11} Nm^{2}kg^{-2}).

Tasks:

(a) Calculate the gravitational force acting on the satellite.

(b) Determine its orbital speed.

(c) Find the total mechanical energy of the satellite in orbit.

(d) Discuss how measurement and dimensional analysis can be used to verify the formula for gravitational potential energy.

Item 35: Projectile Motion and Energy

Topics: Motion under Gravity, Linear Motion, Work/Energy/Power, Fluid Mechanics

A ball is projected horizontally from the top of a building 20 m high with a speed of 15 ms^{-1}.

Tasks:

(a) Calculate the time taken for the ball to reach the ground.

(b) Find the horizontal distance traveled before impact.

(c) Determine the kinetic energy of the ball just before hitting the ground (mass = 0.5 kg).

(d) Discuss how air resistance (fluid mechanics) would affect the motion.

Item 36: Friction and Inclined Plane

Topics: Solid Friction, Statics, Mechanical Properties of Matter

A block of mass 10 kg rests on a rough inclined plane at 30° to the horizontal. The coefficient of static friction is 0.5.

Tasks:

(a) Draw a free-body diagram of the forces acting on the block.

(b) Calculate the maximum frictional force.

(c) Determine whether the block will slide or remain at rest.

(d) Explain how the mechanical properties of matter (surface roughness) influence friction.

Item 37: Work Done by a Pump

Topics: Work/Energy/Power, Fluid Mechanics, Measurement

A water pump raises 200 liters of water per minute from a well 15 m deep. Density of water = 1000 kgm^{-3}.

Tasks:

(a) Calculate the mass of water raised per minute.

(b) Determine the work done by the pump per minute.

(c) If the pump operates at 60% efficiency, calculate the power input required.

(d) Discuss how fluid mechanics principles explain the flow of water through the pump.

Item 38: Rotating Space Station

Topics: Circular Motion, Gravitation, Mechanical Properties of Matter, Measurement & Dimensions

A cylindrical space station of radius 100 m rotates to provide artificial gravity.

Tasks:

(a) Calculate the angular speed required to simulate Earth's gravity.

(b) Determine the linear speed of a person standing at the rim.

(c) Discuss how mechanical properties of matter (structural strength) limit the design.

(d) Explain how measurement and dimensional analysis can verify the formula for centripetal acceleration.

Item 39: Gravitational Potential and Escape Velocity

Topics: Gravitation, Work/Energy/Power, Fluid Mechanics, Circular Motion

A rocket is launched from Earth's surface.

Tasks:

(a) Derive the expression for escape velocity using energy principles.

(b) Calculate the escape velocity from Earth (radius = 6.4×106 m, M=6.0×1024 kg).

(c) Compare the escape velocity with the orbital velocity of a satellite at the same radius.

(d) Discuss how fluid mechanics (air drag) affects the rocket during launch.

Item 40: **Motorcycle Transport on a Hilly Road (Boda-Boda Scenario)**

Topics: Linear Motion, Motion under Gravity, Work, Energy & Power, Friction

A boda-boda rider transports a passenger up a hill inclined at 10° to the horizontal. The combined mass of the rider and passenger is 140 kg. The motorcycle engine delivers a constant power of 2.8 kW while moving at steady speed. The coefficient of friction between the tyres and road is 0.3.

Tasks

a) Determine the resistive force acting on the motorcycle due to friction and component of weight.

b) Calculate the steady speed of the motorcycle up the slope.

c) Explain what would happen to the motion if the rider suddenly reduces engine power.

d) State two safety implications of friction in this situation.

Item 41: **Water Tank Design in a Rural Home**

Topics: Fluid Mechanics, Measurement & Dimensions, Mechanical Properties of Matter

A cylindrical water tank is elevated 5 m above ground level to supply water to a house. The tank is made of steel and holds 2000 litres of water.

Tasks

a) Determine the pressure exerted by the water at the base of the tank.
b) Calculate the force acting on the base of the tank if its radius is 0.5 m.
c) State the SI units and dimensions of pressure.
d) Explain why the tank material must have high tensile strength.
e) Suggest two design improvements to prevent tank failure.

Item 42: Satellite Motion Around Earth

Topics: Circular Motion, Gravitation, Work & Energy

A communication satellite orbits the Earth at a height of 300 km above the surface. The mass of Earth is 6.0×10^{24}kg and its radius is 6.4×10^{6}m.

Tasks

a) Show that gravitational force provides the centripetal force for the satellite.
b) Calculate the orbital speed of the satellite.
c) Determine the total mechanical energy of the satellite.
d) Explain why satellites do not fall to Earth despite gravitational attraction.

Item 43: Lifting Construction Materials Using a Pulley System

Topics: Statics, Work/Energy/Power, Measurement

At a construction site, workers use a pulley system to lift cement bags of mass 50 kg to a height of 8 m. The system operates at 75% efficiency.

Tasks

a) Draw a free-body diagram of the load and identify forces.
b) Calculate the useful work done in lifting the load.
c) Determine the input work required.
d) Calculate the power if the lifting takes 10 seconds.
e) State two sources of energy loss in the system.

Item 44: Vehicle Negotiating a Circular Bend

Topics: Circular Motion, Friction, Gravitation, Linear Motion

A car of mass 1200 kg moves at a constant speed around a horizontal circular bend of radius 50 m. The coefficient of friction between tyres and road is 0.4.

Tasks

a) Calculate the maximum speed at which the car can move without skidding.
b) Explain the role of friction in circular motion.
c) Describe what would happen if the road were wet.
d) Suggest two engineering solutions to improve safety on such bends.
e) State the relationship between centripetal force and linear velocity.

Item 45: Measuring Acceleration of a Rolling Ball

Topics: Measurement & Dimensions, Linear Motion, Motion under Gravity

A student releases a steel ball from rest at the top of a smooth inclined plane and measures the time taken to travel different distances.

Tasks

a) Describe how the student can determine acceleration from the measurements.
b) Derive an equation relating distance, time, and acceleration.
c) State the SI unit and dimensions of acceleration.
d) Explain two possible sources of experimental error.

Item 46: Brick Stability on a Wall

Topics: Statics, Friction, Mechanical Properties of Matter

A brick rests on a rough horizontal surface. A horizontal force is gradually applied until the brick begins to slide.

Tasks

a) Define limiting friction.
b) Determine the frictional force if the coefficient of friction is given.
c) Draw and label a free-body diagram.
d) Explain why wider bases improve stability in construction.

ITEM 47: Water Flow Through a Pipe System

Topics: Fluid Mechanics, Work/Energy, Measurement

Water flows through a horizontal pipe of varying cross-section supplying a household.

Tasks

a) State the principle of continuity.
b) Explain how velocity changes with cross-sectional area.
c) Calculate pressure difference using Bernoulli's principle.
d) State the SI unit and dimensions of flow rate.

Item 48: Lifting Water Using a Pump

Topics: Work, Energy & Power, Motion under Gravity, Fluid Mechanics

A water pump lifts 500 kg of water per minute to a height of 6 m.

Tasks

a) Calculate the work done per minute.
b) Determine the power of the pump.
c) Explain the effect of increasing height on power requirement.
d) Suggest ways to improve pump efficiency.

Item 49: Stretching a Metal Wire

Topics: Mechanical Properties of Matter, Measurement, Work & Energy

A wire is stretched by adding masses and the extension is measured.

Tasks

a) State Hooke's Law.
b) Define Young's modulus.
c) Calculate energy stored in the wire.
d) Explain why elastic limit should not be exceeded.

ITEM 50: Stone Thrown Vertically Upwards

Topics: Linear Motion, Motion under Gravity, Measurement

A stone is thrown vertically upwards with initial velocity 20 m/s.

Tasks

a) Calculate maximum height reached.
b) Determine time of flight.
c) Draw velocity-time graph.
d) State assumptions made in calculations.

ITEM 51: Oil Floating on Water

Topics: Fluid Mechanics, Density, Measurement

Oil is poured into a container of water and forms a layer on top.

Tasks

a) Explain why oil floats on water.
b) Define density and relative density.
c) Calculate density given mass and volume.
d) State units and dimensions of density.

Item 52: Car Braking on a Straight Road

Topics: Linear Motion, Friction, Work & Energy

A car moving at 25 m/s comes to rest after brakes are applied.

Tasks

a) Calculate deceleration if stopping distance is given.
b) Determine braking force.
c) Explain energy transformation during braking.
d) State factors affecting stopping distance.

Item 53: Satellite Escape from Earth

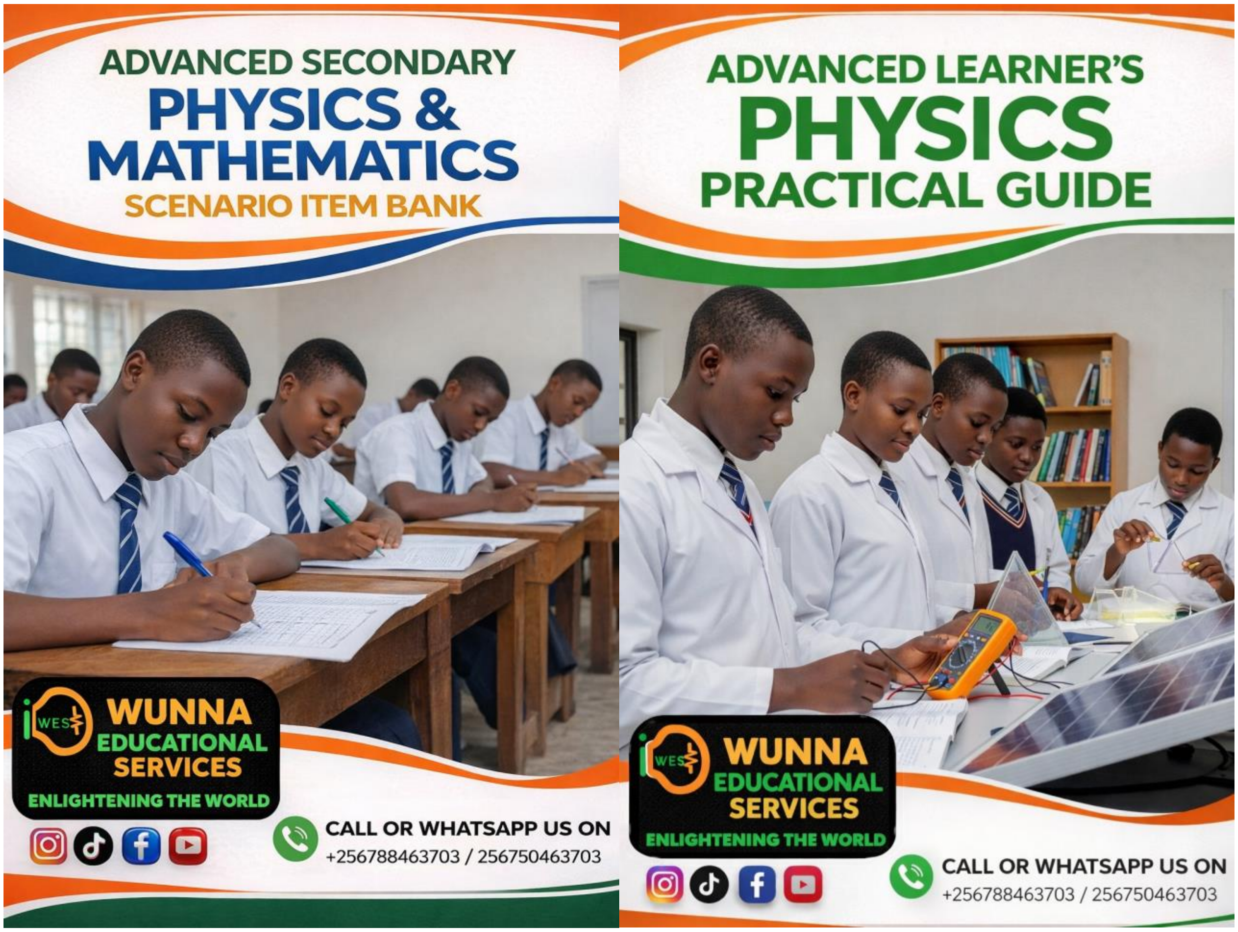
Topics: Gravitation, Work & Energy, Circular Motion

A satellite is projected from Earth's surface.

Tasks

a) Define escape velocity.
b) Derive expression for escape velocity.
c) Calculate escape velocity for Earth.
d) Explain why mass of satellite does not affect escape velocity.

A-LEVEL PHYSICS BOOKS AVAILABLE:



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